



THE UNIVERSITY OF SYDNEY

SUSTAINABLE TEACHING LABS

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EXECUTIVE SUMMARY

The Faculty of Aerospace, Mechanical, and Mechatronics Engineering (AMME) at the University of Sydney faces significant capacity and functionality issues within their first and second-year labs and workspaces in the J07 building, particularly in the S173 Halliday Lab/Makerspace. The primary issues include:

Inadequate Functional Capacity: The current setup does not adequately support the number of students, hindering their ability to gain hands-on experience.

Accessibility: The existing facilities are not fully accessible to students with physical disabilities or neurodiverse students.

Sustainability: The labs are not aligned with the university's Net Zero Initiative, leading to higher carbon emissions and environmental footprints.

Quality of Instruction: There is a need for more experienced demonstrators who can effectively use lab equipment and engage students.

To address these issues, reforms are required that enhance the functional capacity of the labs, ensure accessibility for all students, promote sustainable practices, and improve the quality of instruction. The proposed solution involves a comprehensive remodelling of the existing labs and workspaces to improve utilisation, capacity, sustainability, and accessibility. The key components of the solution include:

Modified HVAC System: An HVAC system will be installed to improve air quality and energy efficiency.

Motion-Sensor Lighting: To reduce energy wastage, motion-sensor lighting systems will be implemented.

Modular and Movable Furniture: Flexible furniture solutions will be introduced to optimise space usage and improve the learning environment.

Accessibility Enhancements: Additional resources such as bone-conduction headphones and tactile signage will be provided to accommodate students with disabilities.

Improved Quality of Instruction: High-quality demonstrators and tutors will be hired and trained to ensure effective teaching and student engagement.

The proposed solution aligns with the engineering requirements of the project, including compliance with the University of Sydney's safety protocols, reduction in emissions and material waste, adherence to budget constraints, and enhancement of accessibility. The design decisions are informed by extensive research and feedback from stakeholders, ensuring that the proposed solutions address the identified problems effectively. The proposed solutions are designed to ensure that the labs and workspaces in the J07 building meet the functional, accessibility, sustainability, and instructional needs of the AMME faculty and students, maintaining the University of Sydney's high teaching and learning standards.

TABLE OF CONTENTS

1. INTRODUCTION

2. PROBLEM DEFINITION

3. PROPOSED SOLUTION

4. SOLUTION DEVELOPMENT

4.1 Modified HVAC System

4.2 Motion-sensored Lighting

4.3 Modular and movable furniture

4.4 Other remodelling features

4.5 Technical Depth

4.6 Design Decisions

5. RISK MANAGEMENT PLAN

6. STAKEHOLDERS MANAGEMENT PLAN

6.1 Introduction

6.1.1 University of Sydney Students

6.1.2 University of Sydney Teaching and Lab Staff

6.1.3 University Administration and Governance Bodies

6.1.4 External Partners and Industry Stakeholders

6.1.5 Community and Public Stakeholders

6.2 Purposes of Engagement

6.2.1 University of Sydney Students

6.2.2 University of Sydney Teaching and Lab Staff

6.2.3 University Administration and Governance Bodies

6.2.4 External Partners and Industry Stakeholders

6.2.5 Community and Public Stakeholders

6.3 Stakeholder Engagement Strategies

6.3.1. Regular Stakeholder Meetings and Workshops

6.3.2. Surveys and Feedback Mechanisms

6.3.3. Stakeholder Advisory Committee

6.3.4. Tailored Communication Channels

6.3.5. Collaborative Decision-Making Processes

6.3.6. Transparent Reporting and Documentation

6.4 Engagement level matrix

6.5 Communication Plan and Engagement timeframe

6.5.1 University of Sydney Students

6.5.2 University of Sydney Teaching and Lab Staff

6.5.3 University Administration and Governance Bodies

6.5.4 External Partners and Industry Stakeholders

6.5.5 Community and Public Stakeholders

6.6 Monitoring and Evaluation

6.6.1 Key Performance Indicators (KPIs)

6.6.2 Monitoring Activities

6.6.3 Evaluation Activities

6.6.4 Reporting and Communication

6.7 Summary of Key Findings

6.7.1 University of Sydney Students

6.7.2 University of Sydney Teaching and Lab Staff

6.7.3 University Administration and Governance Bodies

6.7.4 External Partners and Industry Stakeholders

6.7.5 Community and Public Stakeholders

7. PROJECT IMPLEMENTATION PLAN

8. ECONOMIC PLAN

8.1 - Project Costing

9. CONCLUSION

10. REFERENCES

11. APPENDIX

INTRODUCTION

In its early days, the University of Sydney did not offer an Engineering program, operating as a 'liberal arts' college instead since its establishment in 1850. However, a significant turning point occurred in 1882 with the appointment of Mr. William H. Warren as a Lecturer in Engineering.

The origin of the School of Aerospace, Mechanical and Mechatronics Engineering (AMME) dates back to the year - 1883, where the addition of few professors led to the foundations of mechanical engineering. Later, the Aeronautical/Aerospace came into being with the appointment of Professor A. V. Stephens in the year - 1939 and soon we will be 100 years old in 2039. With time, the number of faculty and enrolled students has only grown and with the emergence of new technologies - a new Mechatronics undergraduate program was also introduced, which was initially termed 'Industrial Automation'. [1][2]

PROBLEM DEFINITION

The problem posed in front of us is the current capacity and functionality problems faced by the Faculty of Aerospace, Mechanical and Mechatronics Engineering (AMME) in the first and second year labs and workspaces in J07 building, especially S173 Halliday Lab/Makerspace. New reforms are therefore needed to increase this functional capacity while ensuring that the hands-on skills relevant to student's discipline are being maintained along with following suitable sustainable practices when in use. This would require deep research and understanding the needs of the stakeholders, especially the AMME which have a direct relation with the students and the learning spaces to come up with possible solutions that can be implemented over time which will cover all the four contexts - legal, policy, social and environmental. Additionally, we have to assure the stakeholders that the policies laid by them are followed at all times while facing these challenges. Further, the necessity to increase the current accessibility to accommodate people of various physical abilities and neurodiverse students becomes pivotal as the number of students increases. Hence, introducing more measures and technologies like bone-conduction headphones and tactile signage to the already existing means like wheelchair accessibility - via the use of ramps or lifts will be effective in enhancing the accessibility. (*The University of Sydney - Sign In*, n.d.)

In lieu of the Net Zero Initiative followed by the University of Sydney, the university aims to reduce the carbon emissions and environmental footprints of the campus to zero by the year 2030. This makes the matter imperative to look after and track for the emissions released by the labs and workspaces while working towards opportunities and ways to reduce them. Furthermore, by meeting people in-person we also got more information in regards to how a ventilation system is required in few of the labs and the dire need for more experienced demonstrators who knows how to use the lab equipment along with an ability to make the tutorials more collaborative and engaging. All the aforementioned points are the key elements being faced by the staff and students alike at the moment, and therefore, a viable solution is required for the betterment of the existing obstacles. This will also ensure that the global teaching and learning standards displayed by the University, which is also a stakeholder of this project, are maintained.

After thinking of various possible ways to address this problem, the most suitable solution that all our team members decided upon after using the decision matrix [\[Appendix 1\]](#) is to remodel the existing labs and workspaces in a more dynamic and aesthetic manner to make the rooms more accommodating and make learning environments more engaging and better for the students. (*The University of Sydney - Sign In*, n.d.)

PROPOSED SOLUTION

The solutions presented in this project aim to help improve the utilisation, capacity and also the sustainability of the targeted space. This includes a brand new floor plan [\[Appendix 5\]](#) of the labs with changes being made in the interior with additional components such as bag hooks, foldable white board tables etc. To tackle present energy wastage, a new motion lighting system [\[Appendix 3\]](#) and an improved version of a HVAC system [\[Appendix 4\]](#) are being implemented shown in the following diagram in the Appendix. Also focussing on the present accessibility issues, we aim to improve this by introducing binary tactile signage, bone conduction headphones and ADHD-friendly resources. Further, we plan to hire high-quality demonstrators and tutors possessing necessary teaching-skills and sufficient hands-on training which will make the learning environment conducive, productive and inclusive for everyone.

SOLUTION DEVELOPMENT

There are few uptakes that we took from the current scenario being faced by the staff and students which together form the base of our proposed solutions. Our solution is mainly comprised of the following components:

4.1 Modified HVAC System

While implementing, there are certain steps that are involved and things that we need to consider, which includes the following:

Before this phase begins, an evaluation of the assessments based on cooling, heating and ventilation requirements in-consideration of factors like efficiency, people in that area, safety regulations etc. Then a detailed report along with a prospective budget is sent to the stakeholders mainly - the University and the AMME faculty to get approval for this project. Once the project gets approved, an order is placed for the HVAC systems and the contractors are made aware of this restoration plan along with the rest of the stakeholders - staff, students and contract labourers. The building is closed to the public while workers assess the results of safety tests to ensure that it is safe to begin removing any existing old parts of HVAC systems in the laboratory or workspace. This ensures that the workers are safe from the electrical and fire hazards alongside risks of electrocution, leaks or breakdowns.

Otherwise, a detailed plan is made by the contractors and architects to layout where the wirings and new HVAC system can be installed while following safety protocols and ensuring that the fundamental structure of the building doesn't get damaged. The next step of the process is to change the old wirings if there's any damage present and then set up the HVAC system in place all over labs and workspaces present in the building. A review about the workings of the newly installed HVAC system based on performance, energy consumption, indoor air quality, etc. and safety checks are carried out until it is free from any public risks before the building is reopened. A normal HVAC system runs based on water, with the new HVAC system running on air based, it removes the need of having extra motor power to push the water. This helps reduce energy usage to the system. The risk of not going beyond the approved financial budget, construction delays, system malfunctions and construction waste as well as the problem of asbestos is consistent through-out the construction process. After conducting thorough research, we have determined that the maximum cost for a single HVAC system is approximately 8,000 AUD. Additionally,

the installation expenses are estimated to be around 12,000 AUD, while hiring construction contractors is expected to amount to 14,000 AUD. The testing and maintenance of the HVAC system is anticipated to cost around 150 AUD and 600 AUD, respectively. [\[Table 1\]\[Appendix 4\]](#)(*The University of Sydney - Sign In*, n.d.)

4.2 Motion-sensored Lighting

Before implementing this part of the solution, we need to notify the stakeholders - the University and the respective faculty (which is AMME) to get their approval for this revampment. After getting the approval, rest of the stakeholders like staff, students and contractors are alerted of this repair work and new lights and a motion sensor system are purchased. Then an evaluation of safety tests is conducted before old lighting and systems are removed. This prevents the contract labourers from the risk of electrical and fire hazards. If there are no wiring damages and issues found, then the new systems are installed otherwise, wiring is changed before the installation. And lastly, new lights are put in place. The additional risks associated with this renovation includes insufficient project funding, system malfunctions and frequent maintenance issues. Furthermore, potential delays or unavailability of labour or contractor services pose a significant risk, along with the potential for construction waste and asbestos-related issues. Through online research, we deduced that the maximum cost for a single motion-sensored lighting unit is approximately 80 AUD and the estimated cost for installation is 500 AUD. [\[Table 1\]\[Appendix 3\]](#)(*The University of Sydney - Sign In*, n.d.)

4.3 Modular and movable furnitures

Before introducing any new furniture to the rooms, we need to identify the areas where flexible furniture could be incorporated. To achieve this, we create a floor plan to see how we can optimise the use of space in the lab. We must carefully consider factors such as ergonomic design, safety, aesthetics, and practicality of use. After personally visiting the Halliday/Makerspace area, we have come up with an exciting idea to replace the current whiteboards with a foldable, magnetically attached option that doubles as a side table. To ensure inclusivity, we have enhanced the foldable whiteboard with a string mechanism that allows for easy accessibility. Not only does this solution provide a modern and stylish touch, but it also considers the needs of all students, including those with disabilities. This innovative solution provides space for students to sit and collaborate on activities. Furthermore, to optimise space and minimise clutter, we plan to install hooks on the walls for students to hang their bags, providing more room to move around. This aligns with the plan created after analysing the new floor plan. By doing so, we create a more functional and aesthetic environment that promotes productivity and engagement. [\[Appendix 5\]](#)(*The University of Sydney - Sign In*, n.d.)

Once the assessment of the area where the new furniture is to be set up is done, we submit a detailed report along with this assessment and its benefits to the stakeholders - University of Sydney and the faculty of AMME. After receiving the approval and financial budget from the aforementioned stakeholders, the order is placed with the university's approved supplier/dealer and the contractor is made aware of this plan. Later, when the furniture is delivered, the contractor and the management finalise a start date for the work and subsequently, the remaining stakeholders - staff and students are informed of this renovation at least a week in advance. Before, the stakeholders - contract workers remove the old goods and replace them with the new ones, a safety check is performed. The associated risks during the

installation of new fittings are: if a detailed research on the various furniture options isn't accomplished then it will hinder the productivity of the students and experiments overall as well as we have to ensure that factors like: ease of movement, durability and compatibility with the experiments being carried out in that room, etc. are considered. If the furniture isn't properly fitted then it will lead to reduced workspace efficiency and might even disrupt the workflow of students. Another risk is the damage to the integrity of the building during the work process and unexpected delays or unexpected delays from the labourer's or contractor's side which might push the project deadline. After conducting thorough research, we have determined that the maximum cost for a single foldable whiteboard desk is approximately 1000 AUD. Additionally, the expenses for the bag hook wall mount and a single unit of bag hooks are estimated to be around 1000 AUD and 10 AUD respectively. [\[Table 1\]](#)(*The University of Sydney - Sign In*, n.d.)

4.4 Other remodelling features

As improving the accessibility for all is of significant importance, we can do this by introducing ADHD-friendly resources, binary signages and bone conduction headphones to make the learning environment more inclusive. The first step of implementation is to plan how many binary tactile signage and bone conduction headphones are required on the basis of the number of people with special needs who have enrolled for the particular course. After gathering all the data, the next step is to inform the university and the faculty of AMME about this plan via a detailed report outlining the expected work-process and financial cost. Once the plan is approved and funding has been secured, the goods will be ordered and upon delivery, the contractor (another stakeholder) will set them up in the labs. The contractor will be informed in advance about this work and staff and students are informed in advance as well if it causes any disruptions to their labs or tutorials. The risk of malfunctioning persists from the time of delivery to the installation phase. Additionally, there is a potential for mishandling by the staff and students, which may result in the frequent disposal of older equipment and the introduction of new ones. (*The University of Sydney - Sign In*, n.d.)

Another feature is to ensure high-quality demonstrators and tutors are hired, and sufficiently incentivized to encourage their long-term commitment. This can be implemented in following ways: first is to inform the faculty of AMME and the university - which are part of the main stakeholders, about this predicament and mutually discuss ways to improve or resolve this problem after receiving feedback from student surveys. Once the suggestions have been met with proper consideration and analysis, the next step is to draft a proposal for the execution of this process. After the proposal and the required funding has been approved, the other stakeholders - staff and students are notified and encouraged to apply for the job. Afterwards, a rigorous selection process for the position begins so as to ensure the individual possesses quality teaching skills and can adapt to new technologies quickly. Thereafter, when the applicants have been shortlisted they are informed of the training sessions which are mandatory to attend and they are provided with the required training to enable them to efficiently handle the lab-work on their own. It is also crucial to ensure that adequate incentives such as competitive hourly wage, performance-based bonuses, staff benefits and recognition and appreciation are provided to retain the high-quality demonstrators and tutors for a long time. The risks associated with this work process are: the potential difficulty to find suitable applicants, it is a challenge to identify individuals who not only excel in conducting experiments but also possess strong communication skills to effectively convey the outcomes to the students., the risk of exceeding the financial constraints set by the university administration. Careful consideration should be given to ensure that expenses remain within the approved limitations. After

extensive research, we have determined that the maximum cost for a single bone conduction headphone set and tactile signage are approximately 270 AUD and 100 AUD. Additionally, the expenses for hiring high-quality demonstrators and tutors amounts to 90,000 AUD per person and training an individual would cost around 5000 AUD. [\[Table 1\]](#)(*The University of Sydney - Sign In*, n.d.)

4.5 Technical Depth

The following statements represents the engineering requirements of our solution:

- All laboratory equipment must comply with the latest 2024 safety protocols as outlined in the University of Sydney's policies.
- Reduce Scope 1 and 2 emissions and material waste by 5 to 10% in the first year (i.e. 2024) of implementing the new spaces/ideas in labs and workspaces.
- The University's allocated budget must be strictly adhered to in accordance with the Building Projects Approval and Management Policy of 2014.
- New spaces must be equipped with appropriate safety measures according to the safety standards established by the University of Sydney.
- Special waste disposal unit must be installed to improve proper disposal of organic, hazardous and plastic waste in each teaching lab.
- Workplace health and safety (WHS) policies and procedures are to be strictly followed at all times during the construction and operation of labs to ensure well-being of everyone involved.
- It is of utmost importance to ensure that any renovations carried out do not compromise the heritage of the J07 building.
- It is critical to satisfy that all equipment and furniture are accessible to students with diverse physical abilities and neurodiversity.
- High-quality demonstrators and tutors must be selected to teach in the session, in accordance to the standards set by University of Sydney.
- Increasing Functional Capacity of the current teaching labs and workspaces.

The proposed solution has been extensively researched using various resources, this can be further understood visually by referring to the Systems Diagram [\[Appendix 2\]](#). For a clear visual representation of our solution, we have included several tables, such as the Decision Matrix [\[Appendix 1\]](#) which presents the initial ideas formulated during the project's planning phase, the Gantt Chart [\[Appendix 6\]](#) that illustrates the expected time management process from the planning stage to the completion of the project, and a Project-Cost chart that provides an estimate for the renovation and repair expenses, among others. (*The University of Sydney - Sign In*, n.d.)

4.6 Design Decisions

As part of remodelling of labs and workspaces, it is necessary to ensure that adequate ventilation and conditioning is provided in all the areas of the building. As per our interactions with a few people in-person, we got to know that the need for HVAC is very much required and hence, included it as part of our solution.

We also found motion-sensored lighting necessary to be included in our solution, as it can alleviate the strain on lab attendants and lecturers who are frequently tasked with turning off lights left on by forgetful individuals. This will also help to reduce energy wastage and will save time at the end of the day.

Introducing new and flexible furniture with a modified floor plan is a vital part of our solution as we saw that the major part of the rooms are cluttered with the workstations and fixed furniture which makes it congested when people are present to do their experiments. As the area of the rooms like S173 Halliday lab is quite small, it would benefit both the staff and students if we incorporate ideas that make the best out of the current space restrictions.

Since, accessibility to all the staff and students is one the primary needs for this project. We kept in mind to work on ideas that cater to them as well. And we came up with the solution of introducing binary tactile signage for the people with sight disability and include bone conduction headphones as well for people with hearing impairments. These can be implemented quite easily as compared to other components of our solution as it will not require any extensive planning, workforce and time.

We also asked some students who had their labs in Halliday lab/Makerspace about what improvements they would like to see in future and the major concern among them was regarding the quality of hired demonstrators and tutors. Acknowledging this matter, we aim to introduce training sessions for all the hired demonstrators and tutors where they can get more hands-on experience and insights into the experiments which will go on across the semester. Providing them with more hands-on experience will enhance the learning environments for students.

All the above feedback that we received during our in-person interactions and the client meetings, has proved invaluable in shaping our design decisions. Through extensive discussions with team members and careful analysis of each of the components, we arrived at these well-considered solutions. (*The University of Sydney - Sign In*, n.d.)

RISK MANAGEMENT PLAN

Risk source/hazard	Likelihood	Potential Impact	Risk Level	Risk Liability	Risk control/mitigation plan	Responsible party for risk control/mitigation
Installation of HVAC System: Construction accidents (electrocution, collapse of concrete),	Possible: During construction, exposure to concrete and electrical system is inevitable.	Minor: Project can be delayed due to injury to worker. New worker needed. Collapse of structure also delay project	Medium	Contractors, the staffs and students.	Safety check list - Electricity is off - Structural strength of building is checked	The employers who are carrying out the job of installation and the university
Installation of Motion Sensor Lighting: In-ceiling installation leads to increased construction accidents (falling, electrocution).	Possible: Exposure to electrical system, installation above ground level	Minor: Injured worker cause delay to project	Medium	Contractors, the staffs and students.	Electricity is off and proper build of scaffold for workers to stand on during installation	The contractors and the university
HVAC system failure: Operational costs for maintenance, no staff & student access to this lab during this time	Likely: Constant running of the system is required hence failure would be expected after some time	Not significant: System failure can be fixed quick.	Medium	The students and staffs	Frequent maintenance and checks on the system	HVAC company and the university
Motion sensor lighting failing: Operational costs for maintenance	Likely: Electrical system is expected to run into problem once or more in their lifespan	Not significant: This can be fixed quickly with appropriate workers	Medium	The students and staffs	Frequent maintenance and checks	Electrician at the University
Fire and Emergency Evacuation Risks: The remodelling of the teaching labs will result in a change of evacuation routes during an emergency or fire. Existing equipment in the current labs and surrounding areas pose as potential flammable sources, and hence, there is a risk posed to the project's implementation.	Possible: The new equipment may impact the evacuation during a fire or emergency but not always.	Moderate: Blockage or obstruction of emergency exit can significantly impact the wellbeing and all beings within the building	High	The university, the students and staffs and all other entity inside the building floor	Clear floor plan with appropriate allocation of equipment. This allows efficient emergency exit path.	The University Environment health and safety department and administration.
Installation and use of new, potentially heavy furniture	Possible: Accident can happen	Minor: Injury to workers or any students/staff can	Medium	The university, staffs, students,	Students and staffs are well educated on how to operate	Contractors and the University

(white board table): The moving and use of flexible furniture into the labs may lead to sprains, strains, fractures and injuries. Through improper lifting or moving, the user is at risk of back injuries.	during relocation of furniture (human error). Accident can also happen during improper use of equipment	occur which will cause delay to the project and cost to the university			safely. Worker should follow safety procedure during moving and installation	
Faulty Bone-conduction earphones: Potential risk of electrocution	Unlikely: Electrocution is possible but unlikely because the headphones must have pass regulation before sold on market.	Minor: Injury to the user will cause minor closure of the facility	Low	The people who install the earphones and students and staffs	Proper User manual for all people. Prevent wrong usage of item	The company who sells the earphones and the university
Compatibility issues: The HVAC system may not be compatible with certain lab equipment, leading to temperature or	Unlikely: This is unlikely because before integrating the HVAC system, there	Minor: This problem can be fixed with the right contractors, does not cause delays or closure of the room	Low	The university, students and staffs	Proper checking before purchasing the System to ensure it is compatible to the building structure	The university and the HVAC company
humidity fluctuations that could affect experiments.	must be a clear evaluation of the system and the structure of the building (including electrical wires etc)					
Not meeting Regulatory/Safety Compliance of HVAC System and Remodelling: Laboratories often have strict regulatory requirements for equipment installation and safety standards. Failure to meet these standards can lead to compliance issues and potential legal consequences	Unlikely: It is unlikely that the HVAC system will not meet safety compliance because laboratory has strict regulatory framework that is controlled under high standards. There are also high stakes for noncompliance	Moderate: Project delay, can also lead to fines and legal action	Medium	The University	By having thorough planning and design, with detailed documentation about the system. Furthermore, having continuous monitoring and quality will ensure the system is meeting Regulatory/Safety compliance.	The University environment and safety department and administration

	ce including fines and legal actions					
University not going ahead with the remodelling plan due to the project costs and remodelling plan	Unlikely: With appropriate budget and feasibility of the project, it is unlikely to be denied	Severe: This will result in the project complete stoppage	Medium	The students, the staffs and other entity using the space that will be renovated	Having statistic that the project is going to bring a good change to the space, improving existing components of the space that improve the wellbeing and study of both students and staff	The project leader, students and staffs are responsible for having this
Hired demonstrators not knowing how to use lab equipment/machinery can cause hazard(s)	Possible: Since there will be new technology and orientation of items in the lab, there are chances that these new demonstrators will need supervision to help operate the area.	Moderate: This risk is moderate because it creates dangerous hazards for both the demonstrator and	High	The students and staffs	Ensure that all new demonstrators undergo direct supervision to operate new machinery within the new space	The existing lab's staffs are responsible for teaching the new demonstrators
Asbestos found within wall lining. (toxic if exposed to)	Unlikely: University rooms and labs are often clean and check. Therefore, it is unlikely to find asbestos	Moderate: Asbestos is highly toxic if exposed to. This may shut down the whole building for further inspection.	Medium	The staffs, students, other departments that is working in the building. Any external working bodies present	Frequent cleaning and checking of building quality, implementation of material that prevent the build-up of asbestos as much as possible.	The university department of health and safety. And external contractor that provide frequent check of building quality and material

Risk Level Matrix

			Potential consequences				
			Minimal consequences. (E.g. Will not significantly delay project, cost implications easy to manage)	Temporary impacts requiring some effort to rectify (E.g. Project delays, deferral of work, require design revisions)	Temporary impact requiring significant effort to rectify	Permanent impact (E.g. requires significant redesign, revision of project scope, requirement to source more funding)	Total loss (E.g. project cancellation)
			Not significant	Minor	Moderate	Major	Severe
Likelihood	Expected to occur regularly under normal circumstances	Almost certain	Medium	High	Very high	Very high	Very high
	Expected to occur at some time	Likely	Medium	High	High	Very high	Very high
	May occur at some time	Possible	Low	Medium	High	High	Very high
	Not likely to occur in normal circumstances	Unlikely	Low	Low	Medium	Medium	High
	Could happen but probably never will	Rare	Low	Low	Low	Low	Medium

STAKEHOLDER MANAGEMENT PLAN

6.1 Introduction

In alignment with the University of Sydney's commitment to excellence in education, research, and sustainability, this stakeholder management plan outlines the strategic approach for upgrading laboratories within the School of Aerospace, Mechanical and Mechatronic Engineering. The primary objective of this initiative is to enhance the learning experience and research capabilities of students and faculty while contributing to the university's broader goal of reducing emissions and environmental impact.

The stakeholders identified for this project include:

6.1.1 University of Sydney Students: Specifically, students enrolled in programs within the School of Aerospace, Mechanical and Mechatronic Engineering. These students are the primary beneficiaries of the laboratory upgrades, as they will directly benefit from improved facilities, equipment, and resources for their academic and research activities.

6.1.2 University of Sydney Teaching and Lab Staff: Faculty members and laboratory technicians play a crucial role in the planning, implementation, and utilisation of upgraded laboratories. Their input, expertise, and support are essential for ensuring that the upgraded facilities meet the needs and expectations of students and align with academic objectives.

6.1.3 University Administration and Governance Bodies: The university administration, including senior management and sustainability officers, are stakeholders in this project due to its alignment with the university's broader goals and commitments. Upgrading laboratories to incorporate energy-efficient technologies and sustainable practices contributes to the university's sustainability agenda, supporting initiatives aimed at reducing emissions and environmental impact campus-wide.

6.1.4 External Partners and Industry Stakeholders: Industry partners, sponsors, and regulatory bodies contribute expertise, resources, and standards relevant to laboratory upgrades and research collaborations. Engaging with these stakeholders ensures alignment with industry best practices and fosters mutually beneficial partnerships.

6.1.5 Community and Public Stakeholders: Local community members and government agencies may be impacted by laboratory upgrades, particularly concerning construction activities and environmental impacts. Engaging with these stakeholders ensures transparency, addresses community concerns, and fosters positive relationships with external stakeholders.

By actively engaging with these stakeholders throughout the project lifecycle, we aim to build consensus, foster collaboration, and maximise the positive impact of laboratory upgrades on student learning outcomes, research excellence, and environmental sustainability. This stakeholder management plan serves as a roadmap for effective communication, consultation, and stakeholder engagement, ensuring the success and sustainability of the laboratory upgrade initiative at the University of Sydney.

6.2 Purposes of Engagement

Engaging stakeholders in the laboratory upgrade initiative at the University of Sydney serves several key purposes, aligning with the project's objectives and the broader goals of the university. The purposes of engagement with specific stakeholder groups are as follows:

6.2.1 University of Sydney Students:

- *Enhancing Learning Experience:* By engaging students in the decision-making process, we aim to ensure that laboratory upgrades meet their educational needs and preferences, enhancing their learning experience and academic success.
- *Gathering Feedback:* Students' feedback and input are essential for identifying areas for improvement and implementing changes that address their concerns and enhance the usability and effectiveness of laboratory facilities.
- *Building Ownership and Pride:* Engaging students in the upgrade process fosters a sense of ownership and pride in their learning environment, motivating them to take ownership of their academic journey and contribute positively to the university community.

6.2.2 University of Sydney Teaching and Lab Staff:

- *Utilising Expertise and Input:* Engaging teaching and lab staff in decision-making processes ensures that their expertise and insights are leveraged to inform the design and implementation of laboratory upgrades, optimising functionality and usability.
- *Securing Support and Collaboration:* By involving faculty members and technicians from the School of Aerospace, Mechanical and Mechatronic Engineering, we aim to secure their support and collaboration, facilitating the successful implementation and utilisation of upgraded laboratories.
- *Promoting Innovation and Excellence:* Engaging teaching and lab staff in discussions around equipment selection, layout design, and pedagogical approaches encourages innovation and excellence in teaching and research, driving continuous improvement and academic excellence within the school.

6.2.3 University Administration and Governance Bodies:

- *Advancing Strategic Priorities:* Engaging with university administration and governance bodies ensures alignment with strategic priorities, securing necessary resources, and ensuring compliance with institutional standards and policies.
- *Demonstrating Accountability:* Transparent and inclusive engagement with university leadership demonstrates accountability and commitment to achieving project objectives while contributing to broader institutional goals and objectives.

6.2.4 External Partners and Industry Stakeholders:

- *Leveraging Expertise and Resources:* Engaging external partners and industry stakeholders allows us to leverage their expertise, resources, and networks to inform laboratory upgrades, foster collaboration, and ensure alignment with industry best practices and standards.

- *Building Sustainable Partnerships:* By involving external stakeholders in decision-making processes, we aim to build sustainable partnerships that extend beyond the university, creating opportunities for knowledge exchange, innovation, and collective impact.

6.2.5 Community and Public Stakeholders:

- *Addressing Community Concerns:* Engaging with community and public stakeholders ensures transparency, addresses community concerns, and fosters positive relationships, contributing to social responsibility and sustainable development goals.
- *Enhancing Stakeholder Trust:* Transparent and inclusive engagement with community and public stakeholders builds trust, enhances credibility, and strengthens the university's reputation as a responsible and engaged member of the community.

6.3 Stakeholder Engagement Strategies

Effective stakeholder engagement is essential for the success of the laboratory upgrade initiative at the University of Sydney. To ensure that the needs, expectations, and perspectives of all stakeholders are considered throughout the project lifecycle, the following engagement strategies will be implemented:

6.3.1. Regular Stakeholder Meetings and Workshops:

- Scheduled meetings will be held with representatives from the School of Aerospace, Mechanical and Mechatronic Engineering, including faculty members, laboratory technicians, and student representatives.
- Workshops will be conducted to gather input on specific aspects of the laboratory upgrades, such as equipment selection, layout design, and sustainability features.

6.3.2. Surveys and Feedback Mechanisms:

- Surveys will be distributed to students, faculty, and staff to gather feedback on their current experiences with laboratory facilities and their preferences for improvements.
- Feedback mechanisms, such as suggestion boxes and online forums, will be established to provide stakeholders with opportunities to share their thoughts, concerns, and suggestions throughout the project.

6.3.3. Stakeholder Advisory Committee:

- A stakeholder advisory committee will be formed, comprising representatives from the School of Aerospace, Mechanical and Mechatronic Engineering, university administration, and relevant external stakeholders (e.g., industry partners, sustainability experts).
- The committee will meet regularly to review project progress, provide guidance and recommendations, and address any challenges or concerns that arise.

6.3.4. Tailored Communication Channels:

- Communication channels will be tailored to the preferences and needs of different stakeholder groups.
- For students, communication may include email updates, announcements in lectures, and social media engagement.

- Faculty and staff may prefer face-to-face meetings, departmental newsletters, and dedicated online portals for project information and updates.

6.3.5. Collaborative Decision-Making Processes:

- Decision-making processes will be inclusive and collaborative, involving stakeholders in key decisions related to laboratory design, equipment procurement, and sustainability initiatives.
- Stakeholders will be invited to participate in focus groups, design charrettes, and review sessions to provide input and feedback on proposed solutions and alternatives.

6.3.6. Transparent Reporting and Documentation:

- Project progress, decisions, and outcomes will be transparently documented and reported to stakeholders.
- Regular progress reports will be distributed via email, posted on project websites, and presented at stakeholder meetings to keep stakeholders informed and engaged throughout the project lifecycle.

By implementing these stakeholder engagement strategies, we aim to foster collaboration, build trust, and ensure that the laboratory upgrade initiative meets the needs and expectations of all stakeholders while contributing to the University of Sydney's goals of academic excellence and environmental sustainability.

6.4 Engagement level matrix

		Power/Influence	
		Low	High
Interest	High	<i>Keep Informed</i> Community and public Stakeholders	<i>Full engagement</i> University of Sydney Students University of Sydney Teaching and Lab Staff University Administration and Governance Bodies
	Low	<i>Little effort required</i>	<i>Keep satisfied</i> External Partners and Industry Stakeholders

6.5 Communication Plan and Engagement timeframe

The engagement time frame outlines when and how often different stakeholders will be engaged throughout the laboratory upgrade initiative at the University of Sydney. This ensures continuous communication, feedback, and involvement at various stages of the project.

6.5.1 University of Sydney Students

Pre-Project Planning

- Initial surveys and feedback collection to understand student needs and preferences.

Project Kick-off

- Kick-off event and initial briefings.

Design and Development Phase

- Monthly email updates.
- Quarterly focus groups and workshops to review plans and gather feedback.

Implementation Phase

- Regular updates via email and social media.
- Bi-monthly meetings to discuss progress and address concerns.
- Hands-on workshops for students to familiarise them with new equipment.

Post-Implementation

- Follow-up surveys to gather feedback on the new facilities.
- Workshops on utilising new technologies and resources.

6.5.2 University of Sydney Teaching and Lab Staff

Pre-Project Planning

- Consultative meetings to gather input on design and equipment needs.

Project Kick-off

- Initial briefings and collaborative planning sessions.

Design and Development Phase

- Monthly email updates and internal newsletters.
- Quarterly workshops to review progress and gather detailed feedback.

Implementation Phase

- Regular updates and bi-monthly departmental meetings.
- Training sessions for staff on new equipment and lab protocols.

Post-Implementation

- Follow-up meetings to evaluate the effectiveness of upgrades.
- Continuous professional development workshops.

6.5.3 University Administration and Governance Bodies

Pre-Project Planning

- Strategic meetings to align project goals with institutional priorities.

Project Kick-off

- Presentation of project plans and objectives.

Design and Development Phase

- Quarterly executive briefings and detailed reports.

Implementation Phase

- Regular updates through formal reports and executive meetings.
- Bi-annual strategic reviews to assess project alignment with university goals.

Post-Implementation

- Final project review and impact assessment.
- Reports on outcomes and strategic benefits.

6.5.4 External Partners and Industry Stakeholders

Pre-Project Planning

- Initial outreach to establish partnerships and gather input.

Project Kick-off

- Briefings to outline project scope and partnership opportunities.

Design and Development Phase

- Bi-annual advisory board meetings to review project progress and provide expertise.

Implementation Phase

- Semi-annual updates and workshops focused on industry standards and best practices.
- samsam

Post-Implementation

- Final consultations to assess project outcomes and potential future collaborations.

6.5.5 Community and Public Stakeholders

Pre-Project Planning

- Community meetings to introduce the project and gather initial feedback.

Project Kick-off

- Public forums to outline project details and expected impacts.

Design and Development Phase

- Quarterly public meetings and updates.
- Regular newsletters and community bulletins.

Implementation Phase

- Continuous communication through local media and community liaison officers.
- Bi-annual public forums to discuss progress and address community concerns.

Post-Implementation

- Community feedback sessions to evaluate the impact of the project.

- Final public meeting to discuss outcomes and future engagement.

By adhering to this structured engagement timeframe, we ensure that all stakeholders are continuously informed, involved, and able to contribute to the successful completion of the laboratory upgrade initiative.

6.6 Monitoring and Evaluation

6.6.1 Key Performance Indicators (KPIs)

The following KPIs will be used to monitor and evaluate the project:

Project Milestones and Timelines

- Percentage of project milestones achieved on schedule.
- Number of delays and reasons for any deviations from the timeline.

Stakeholder Engagement

- Number of stakeholder meetings, workshops, and consultations held.
- Level of stakeholder participation and feedback received.
- Stakeholder satisfaction ratings from surveys and feedback forms.

Quality and Usability of Upgrades

- Compliance with design specifications and quality standards.
- User satisfaction ratings (students, teaching staff, lab staff) regarding new facilities and equipment.
- Number of reported issues or malfunctions with upgraded facilities.

Impact on Learning and Teaching

- Improvement in student performance and learning outcomes related to lab work.
- Faculty and staff feedback on the effectiveness of new equipment and facilities in teaching and research.
- Utilisation rates of the upgraded laboratories.

Sustainability and Environmental Impact

- Reduction in energy consumption and emissions resulting from the upgrades.
- Compliance with sustainability goals and environmental standards.
- Community feedback on environmental impacts and sustainability measures.

6.6.2 Monitoring Activities

To systematically track these KPIs, the following monitoring activities will be conducted:

Regular Progress Reviews

- Monthly project team meetings to review progress against timelines and milestones.
- Quarterly reviews with key stakeholders to assess engagement activities and gather feedback.

Surveys and Feedback Mechanisms

- Periodic surveys distributed to students, staff, and community members to collect feedback on engagement activities and the quality of the upgrades.
- Continuous feedback channels such as suggestion boxes, dedicated email addresses, and online forms.

Site Inspections and Quality Audits

- Regular site inspections to ensure compliance with design specifications and quality standards.
- Quality audits conducted at key stages of the project to identify and address any issues promptly.

Environmental Monitoring

- Ongoing monitoring of energy usage and emissions to measure the environmental impact of the upgrades.
- Environmental impact assessments conducted at the beginning and end of the project to evaluate sustainability measures.

6.6.3 Evaluation Activities

Evaluation activities will focus on assessing the overall success of the project and identifying lessons learned:

Mid-Project Evaluation

- A comprehensive evaluation conducted halfway through the project to assess progress, stakeholder engagement, and preliminary impacts.
- Adjustments to strategies and plans based on mid-project findings.

Post-Implementation Evaluation

- A final evaluation conducted upon project completion to assess the overall impact, stakeholder satisfaction, and achievement of project objectives.
- Detailed reports documenting project outcomes, successes, challenges, and lessons learned.

Stakeholder Feedback Sessions

- Focus groups and feedback sessions with key stakeholder groups to gather in-depth insights and qualitative feedback.
- Incorporation of stakeholder suggestions into final evaluation reports and future project planning.

6.6.4 Reporting and Communication

Monthly Progress Reports: Detailed reports on project progress, stakeholder engagement activities, and any issues or risks identified. Distributed to the project team and key stakeholders.

Quarterly Stakeholder Updates: High-level summaries of progress, key achievements, and upcoming activities shared with all stakeholders.

Final Project Report: Comprehensive report detailing the outcomes, impacts, and lessons learned from the project. Shared with the university administration, governance bodies, and public stakeholders.

6.7 Summary of Key Findings

6.7.1 University of Sydney Students

High Level of Interest and Influence: Students, particularly those in the School of Aerospace, Mechanical, and Mechatronic Engineering, have a significant interest in the laboratory upgrades due to the direct impact on their learning experiences and academic outcomes.

Engagement Strategies: Regular updates, focus groups, and hands-on workshops will ensure continuous student involvement and feedback.

6.7.2 University of Sydney Teaching and Lab Staff

High Level of Interest and Influence: Teaching and lab staff are crucial stakeholders due to their direct interaction with the laboratory facilities and their role in delivering quality education and research.

Engagement Strategies: Consultative meetings, professional development workshops, and ongoing training sessions will ensure their input is integrated and their needs are met.

6.7.3 University Administration and Governance Bodies

High Level of Interest and Influence: These bodies have strategic oversight and are responsible for aligning the project with the university's broader goals and resource allocation.

Engagement Strategies: Strategic briefings, executive reports, and continuous alignment reviews will secure their support and oversight.

6.7.4 External Partners and Industry Stakeholders

Variable Level of Interest and Influence: While their involvement is valuable for expertise and industry standards, their interest may be more limited compared to internal stakeholders.

Engagement Strategies: Targeted outreach, advisory board meetings, and alignment with industry standards will maintain their engagement.

6.7.5 Community and Public Stakeholders

High Level of Interest and Influence: Community and public stakeholders exhibit a high level of interest due to the environmental impacts and their expectations for responsible development.

Engagement Strategies: Public forums, environmental impact assessments, and continuous transparency will address their concerns and foster positive relations.

PROJECT IMPLEMENTATION PLAN

This Gantt chart [\[Appendix 6\]](#) provides a comprehensive timeline for the project, capturing all the phases from initial planning to the final implementation of the solution which aimed to enhance the lab capacity and sustainability. The planning stage begins with meeting with clients, following up by identifying the key problems and their solutions. Furthermore, it is followed with stakeholders' management plan, risk management plan and establishing the project budget.

The charts provide further details on the timeline to secure needed funding and resources to ensure all the support and products are in place for the project's success. It is followed by the construction phase, which includes hiring outside contractors and a risk management plan for each construction phase. Milestones are highlighted for each construction phase; this provides greater understanding of the deadline such that the whole project can proceed smoothly without disruption. The key solutions were implementing energy efficient systems such as the new improved HVAC system [\[Appendix 4\]](#) and the motion lighting system [\[Appendix 3\]](#). The chart shows critical phases like the project approval and quality check and inspection to ensure all compliance and quality of the solutions are met. Throughout the project, the Gantt chart [\[Appendix 6\]](#) plays a key role as a visual aid to monitor the process and allocate budget and resources efficiently such that the project can proceed. With a clear outline and time, it ensures that the project stays on track and meets the deadline.

ECONOMIC PLAN

8.1 - Project Costing

Table 1: Project Costing Table showcasing the various phases, costs, and revenue associated with the proposed solution.

Phase	Item/Task	Unit cost		Units		Total		Income	References
		Upper	Lower	Upper	Lower	Upper	Lower		
Project Planning	Provided Budget from USYD	0	0	0	0	0	0	1000000	
	HVAC System	8500	1600	2	1	17000	1600		(AUSAIR Conditioners, 2022)
	Motion Sensored Lighting	78	56	10	7	780	392		(Lighting Style, 2024)
	Foldable Whiteboard Desk	1000	800	24	12	24000	9600		(WRITEY, 2024)
	Bag Hooks Wall Mount	1000	790	4	2	4000	1580		(Safehook Australia, 2024b)
	Bag Hooks	10	7	120	60	1200	420		(Safehook Australia, 2024a)
	Bone Conduction Headphones	270	50	50	30	13500	1500		(Hiteblaz, 2024)
	Tactile Signage	100	50	5	2	500	100		(NewSigns, 2024)
	Hired Demonstrators	90000	70000	4	1	360000	70000		(talent.com, 2024)
Implementation	Hiring Construction Contractors	14000	9800	4	2	56000	19600		(indeed, 2024)
	Installation of HVAC	12000	6000	2	1	24000	6000		(hipages, 2023)
	Installation of Motion Sensor Lights	500	350	2	1	1000	350		(Graham, 2023)
	Training Lab Demonstrators	5000	0	4	1	20000	0		(Tafe NSW, 2024)
Reviewing	Testing of HVAC	150	100	2	1	300	100		(Huntercon, 2023)
	Maintenance of HVAC	600	250	2	1	1200	250		(Huntercon, 2023)

Through extensive research, various cost estimations were obtained for the numerous items and tasks required to solve the clients' needs for sustainable teaching labs. Table 1 presents these costs along with the income/budget provided by the University of Sydney. The project phases range from project planning, implementation, and reviewing, and each stage has its own associated costs. Numerous product distributing companies were located to identify the costs of each required item, such as foldable whiteboard desks, bag hooks, HVAC system, tactile signage, and more. Through obtaining the upper and lower bounds of these costs and number of units, the absolute maximum and minimum costing for this project can be achieved.

As previously discussed, the main funding source of this project is the budget provided by the University of Sydney of approximately \$1,000,000. There is a lack of other revenue streams, as this project is intended for use by first-year engineering students, who will pay for the unit courses, which in turn pays for their access to these facilities. As this is not a direct source of funding or revenue for the project, it was not included. Other less prominent **revenue streams** could be from granting agencies, industry agreements or other private foundations, (Holbrook & Sanberg, 2013). Additionally, the main objectives of this project were to increase accessibility to the teaching labs, rather than generate revenue from students. Moreover, the only revenue stream required would be funding from the University of Sydney, and for this to occur, an extensive cost analysis, risk management, and implementation plan must be provided.

Table 1's costing plan also accounts for the **risks** and **uncertainties** identified within the risk management plan, as the system testing and maintenance costs for the HVAC system are included. It was concluded in the risk register that HVAC system failure is probable, and hence, initial and regular system and safety testing of the device will be required. HunterCON Electrical and Air are fully licensed specialists in this

field, and they provided the estimated costs of \$100 to \$150 quotes for testing (Huntercon, 2023). This company is highly suitable for fulfilling the project's solution outcomes due to its location, low costs, and reliability. The risks of hired demonstrators not knowing how to use the existing lab equipment are also accounted for in this costing table. By providing the demonstrators with relevant training courses, this risk is minimised and well-considered. This training was found to range in cost as some TAFE NSW courses could be free depending on the individuals hired (Tafe NSW, 2024).

Table 2: Cumulative Totals calculated from each phase costs

	Phase Cumulative Total		
	Upper	Lower	Income
Project Planning	\$ 420,980	\$ 85,192	\$ 1,000,000
Implementation	\$ 521,980	\$ 111,142	\$ 1,000,000
Reviewing	\$ 523,480	\$ 111,492	\$ 1,000,000

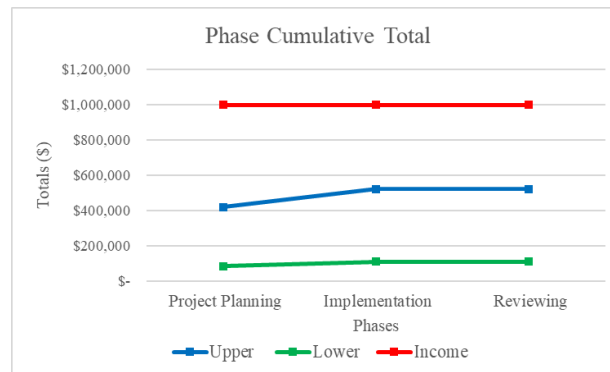


Figure 1: Graph representing the maximum and minimum total costs of the project against the projected revenue

In order to determine the **financial feasibility** of the solution proposed, a phase cumulative total table and graph were produced. The values for these totals are shown in Table 2, and are represented graphically in Figure 1. Through these, it can be concluded that the project is feasible both at its projected maximum and minimum costing. This is due to the much lower total for costs in comparison to the funding/income amount of \$1,000,000. The budget allocated by the University of Sydney for this project is well beyond the budget required, and the cost differences can hence be utilised for other crucial university sectors, such as research facilities, equity support, other teaching spaces, and more (Scott, 2024).

Overall, the estimated costs and income of the solution proposed in this project are deemed to be viable and economically feasible. The difference between the income and maximum costs is around \$477,000, as visualised in the large vertical space between the lines in Figure 1, and this also implies the project's sustainability. Future costs that are not included in this project such as additional system breakages, and sensor light replacements could be accounted for with this additional funding. Ultimately, the cost analysis suggests that the upgrades made to the teaching labs are viable, sustainable, and financially feasible.

CONCLUSION

In conclusion, this project report thoroughly examines the problem delivered by our client to renovate and improve the existing learning space (S173 Halliday lab/Makerspace). Knowing that the population of student's enrollment will increase due to rising interest in engineering studies, it is crucial to expand the capacity and optimise the existing lab spaces and new spaces. In addition, implementation of new systems and improving of existing systems is essential for the long-term sustainability and efficiency of the labs. Many different solutions ideas were discussed in this report with appropriate consideration regarding risks management, stakeholders and project costing. These solutions will be carried out based on the timestamp shown in our Gantt chart. It was made to ensure all actions are up to date and according to plan, as this prevents any clog up during the implementation of our solutions. Introducing exciting ideas like foldable whiteboard desks, binary tactile signage, ADHD-related resources, bone conduction headphones and selecting high-quality demonstrators and tutors with essential teaching-skills and hands-on experience would create an environment that is not only conducive and productive, but also well-organised and inclusive for everyone. Further, implementing a new modified HVAC System [\[Appendix 4\]](#) and motion-sensored lighting [\[Appendix 3\]](#) would ensure that only necessary energy is being consumed which is in lieu with the Net-Zero Emissions target.

Due to some limitations set out by our clients such as the external structure must be kept the same, this limits the variety of solutions that we could have. Therefore, existing solutions mentioned in this report can only do so much to improve the capacity, efficiency and sustainability of the space. Some future recommendations that could better improve the sustainability and capacity of the space are reconstructing the whole floor. This includes having a brand new floor plan, implementing latest technology and optimising all empty space.

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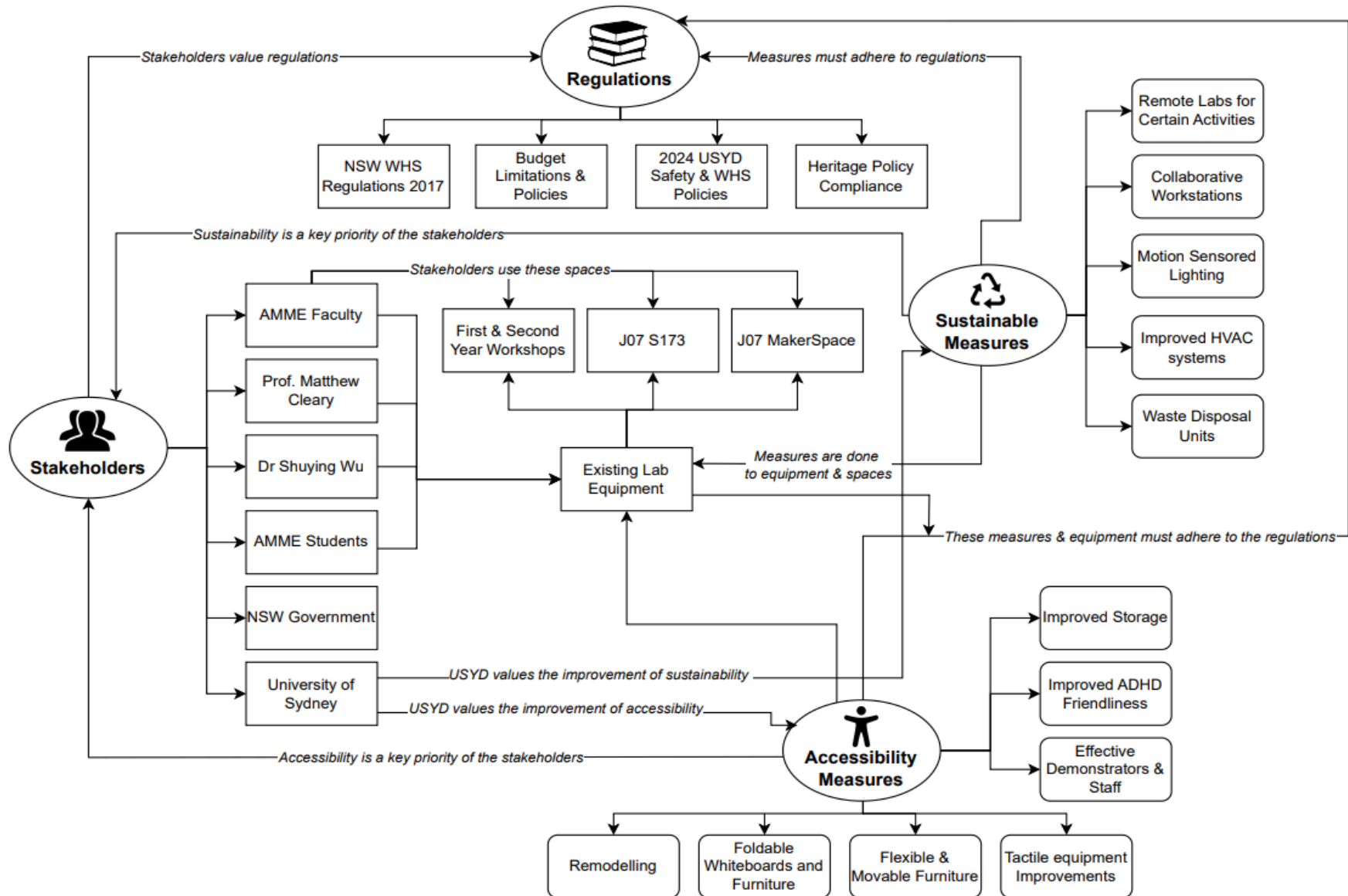
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APPENDIX (BELOW)

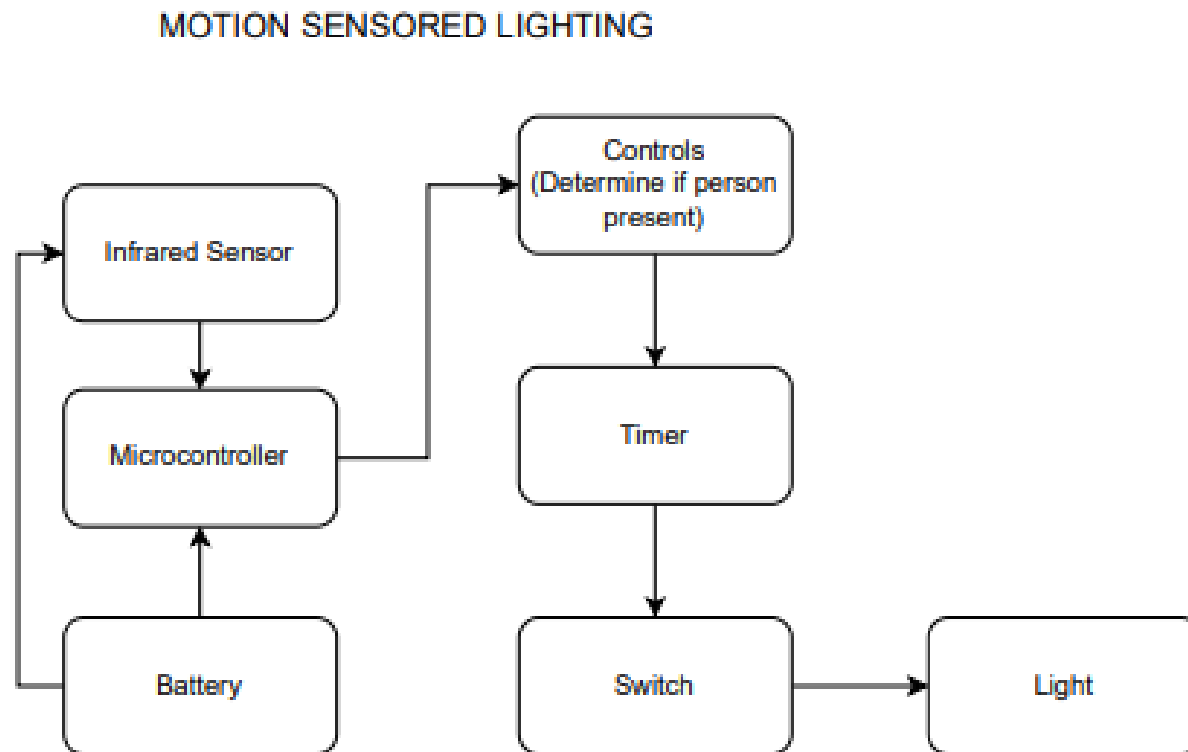
APPENDIX 1: DECISION MATRIX

Engineering Requirements	Criteria	Expansion of Space		Virtual Simulation		Smart Laboratory Technologies		Collaborative Work		
		Weighting	Score	Weighted score	Score	Weighted score	Score	Weighted score	Score	Weighted score
Client priority	Sustainable labs	6	2	12	4	24	7	42	4	24
	Increased capacity of the teaching labs and workspaces	7	7	49	4	28	7	49	7	49
	Accessibility for different physical or neurodiverse students	5	2	10	5	25	7	35	5	25
Problem definitions	Lack of accessibility for students with diverse physical abilities and neurodiversity	5	3	15	6	30	8	40	5	25
	Lack of Sustainability of the teaching labs & workspaces	6	5	30	8	48	8	48	5	30
	Increase in enrolled students	8	5	40	4	32	5	40	8	64
	Lack of functional capacity in workshops and labs	8	5	40	4	32	9	72	5	40
Social context	Enhance accessibility for students with diverse physical abilities and neurodiversity	6	4	24	6	36	8	48	5	30
	Maintain wellbeing of students and teachers in class	9	1	9	5	45	8	72	6	54
Legal Context	According to WHS Government Regulations, maintain wellbeing of staff & students	9	2	18	5	45	8	72	8	72
	Maintaining heritage of the building	7	4	28	10	70	7	49	10	70
	Faculty to strictly follow the budget allocated by the University	8	2	16	8	64	7	56	9	72
Policy context	Accessibility for students with diverse physical abilities and neurodiversity	6	4	24	6	36	8	48	7	42
	Maintain WHS policies of individuals during construction and teaching	9	6	54	5	45	3	27	7	63
	Maintaining heritage of the building as per Uni policy	8	5	40	10	80	8	64	10	80
	New spaces equipped with safety measures as per standards laid by USYD	8	5	40	7	56	8	64	3	24
	Finding high-quality demonstrators as per UniSyd standards	9	1	9	6	54	5	45	6	54
Environmental context	Special waste disposal units should be used	7	2	14	7	49	5	35	3	21
	Reduction of emissions & waste	8	1	8	8	64	8	64	7	56
	Equipments in line with safety & sustainability protocols	6	2	12	7	42	8	48	4	24
	Follow Net Zero Initiative	7	2	14	9	63	8	56	6	42
Team Skills Base	Extensive researching skills	8	6	48	7	56	7	56	7	56
	Teamwork/Collaboration	8	6	48	8	64	9	72	10	80
	Communication	8	6	48	7	56	9	72	9	72
	Construction skills	6	8	48	6	36	6	36	5	30
	Professional Skills	6	5	30	5	30	7	42	7	42
	Adaptability	6	6	36	8	48	7	42	6	36
SCORE				764		1258		1394		1277

APPENDIX 2: SYSTEMS DIAGRAM

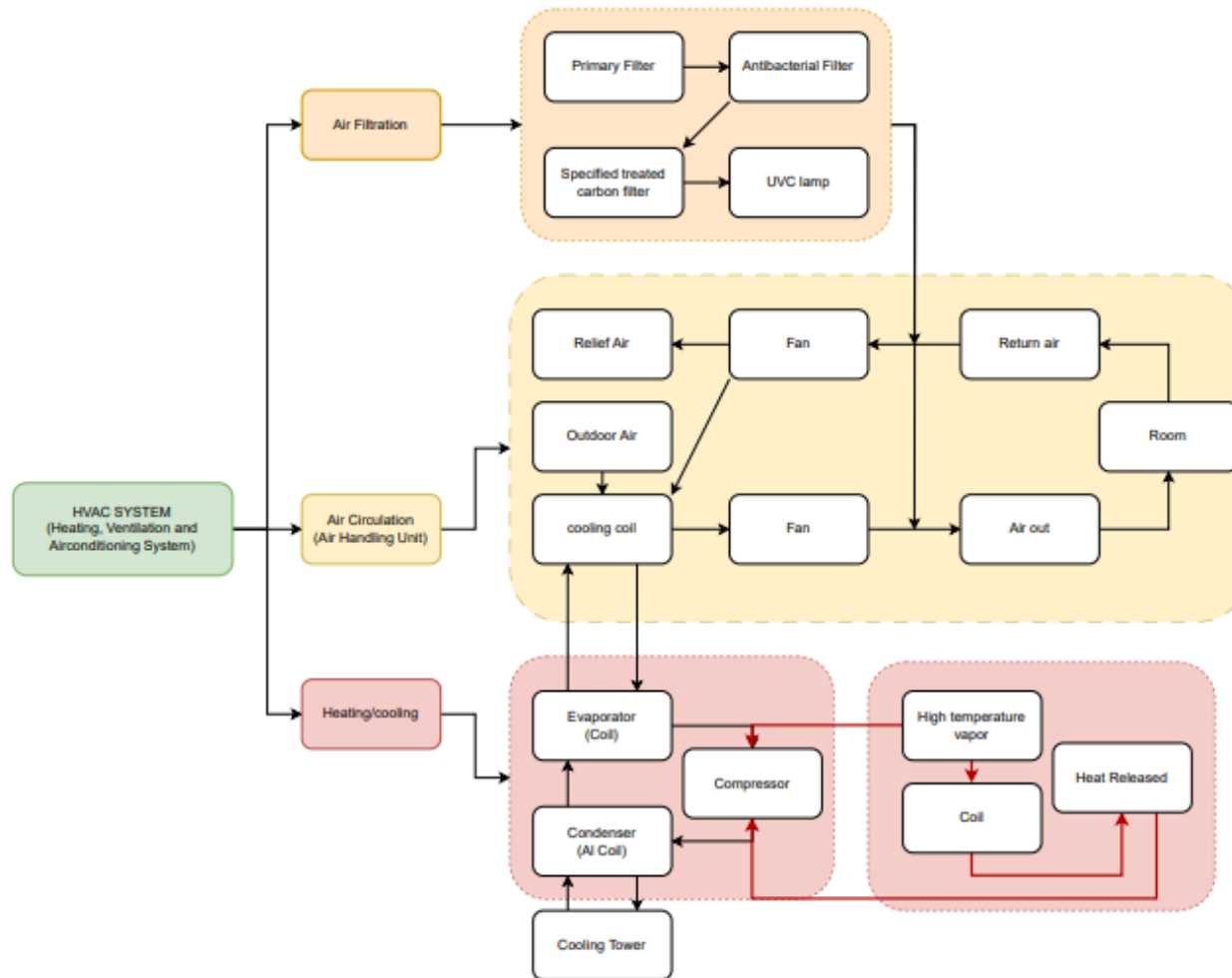


APPENDIX 3: MOTION-SENSORED LIGHTING BLOCK DIAGRAM

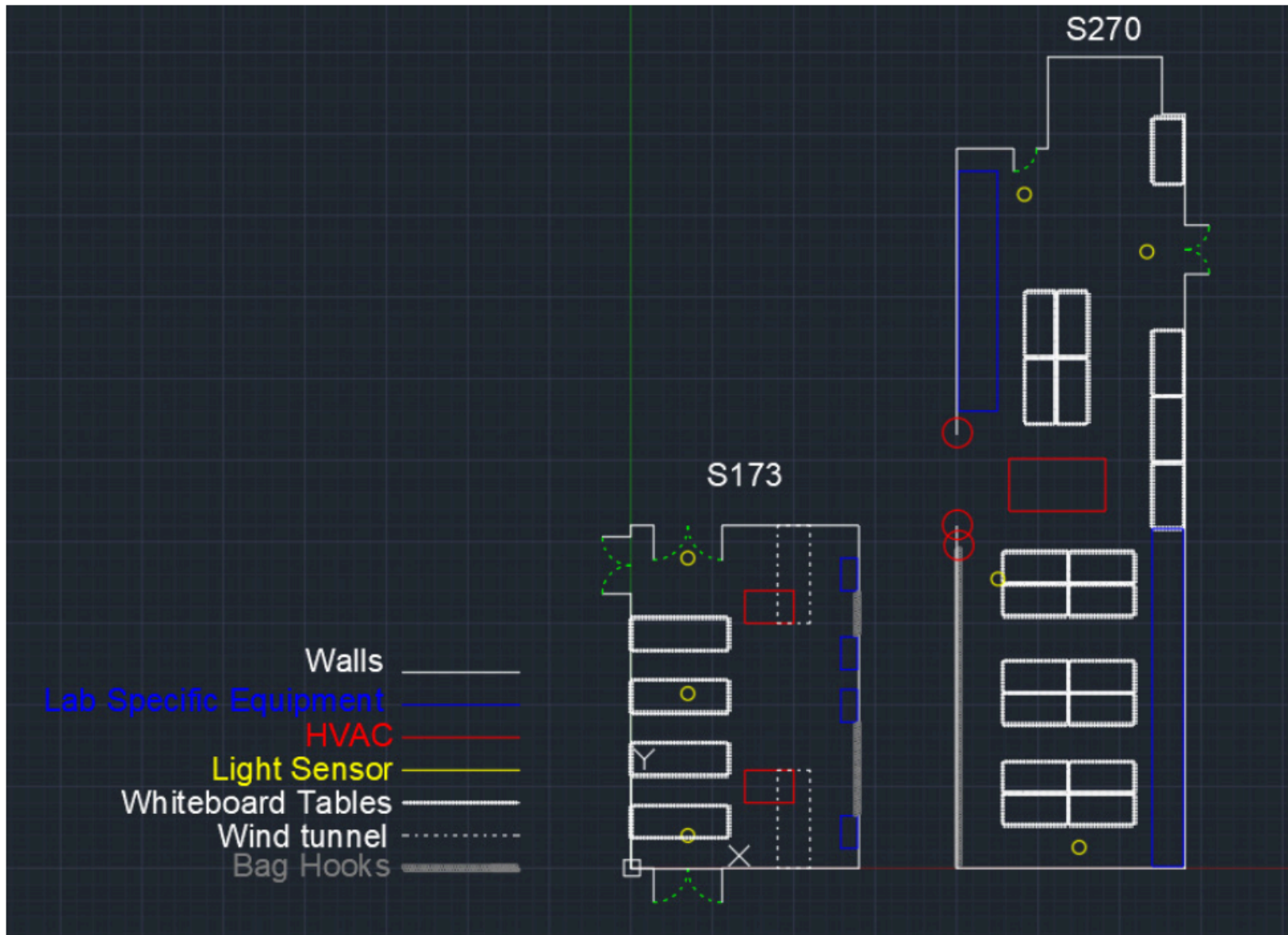


APPENDIX 4: MODIFIED HVAC SYSTEM BLOCK DIAGRAM

HVAC SYSTEM



APPENDIX 5: UPDATED FLOOR PLAN FOR S173 HALLIDAY'S LAB/ S270 MAKERSPACE



APPENDIX 6: GANTT CHART